

Identical Particles

Quantum Mechanics

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1 Two-Particle Systems

For a two-particle system, the wave function depends on coordinates of both particles and time: $\Psi(\mathbf{r}_1, \mathbf{r}_2, t)$. The Schrödinger equation is:

$$i\hbar \frac{\partial \Psi}{\partial t} = H\Psi, \quad H = -\frac{\hbar^2}{2m_1} \nabla_1^2 - \frac{\hbar^2}{2m_2} \nabla_2^2 + V(\mathbf{r}_1, \mathbf{r}_2, t). \quad (1)$$

The probability density is $|\Psi(\mathbf{r}_1, \mathbf{r}_2, t)|^2 d^3r_1 d^3r_2$, with normalization:

$$\int |\Psi(\mathbf{r}_1, \mathbf{r}_2, t)|^2 d^3r_1 d^3r_2 = 1. \quad (2)$$

For time-independent potentials, stationary states are $\Psi = \psi(\mathbf{r}_1, \mathbf{r}_2) e^{-iEt/\hbar}$.

1.1 Center of Mass Separation

For potentials depending only on $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$, define center of mass:

$$\mathbf{R} = \frac{m_1 \mathbf{r}_1 + m_2 \mathbf{r}_2}{m_1 + m_2}, \quad \mu = \frac{m_1 m_2}{m_1 + m_2} \text{ (reduced mass)}.$$

The Schrödinger equation separates:

$$-\frac{\hbar^2}{2(m_1 + m_2)} \nabla_R^2 \psi_R - \frac{\hbar^2}{2\mu} \nabla_r^2 \psi_r + V(\mathbf{r}) \psi_r = E \psi_R \psi_r, \quad (3)$$

with $E = E_R + E_r$. The center of mass moves as a free particle; the relative motion reduces to a one-body problem.

1.2 Bosons and Fermions

For identical particles, the Hamiltonian is symmetric under exchange. The exchange operator P satisfies $P^2 = 1$ with eigenvalues ± 1 . The wave function must be either symmetric (bosons) or antisymmetric (fermions):

$$\psi(\mathbf{r}_1, \mathbf{r}_2) = \pm \psi(\mathbf{r}_2, \mathbf{r}_1). \quad (4)$$

For two particles in states ψ_a and ψ_b :

$$\psi_{\pm}(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{\sqrt{2}} [\psi_a(\mathbf{r}_1) \psi_b(\mathbf{r}_2) \pm \psi_b(\mathbf{r}_1) \psi_a(\mathbf{r}_2)]. \quad (5)$$

The connection between spin and statistics: particles with integer spin are bosons (symmetric); particles with half-integer spin are fermions (antisymmetric).

1.3 Pauli Exclusion Principle

For identical fermions, if $\psi_a = \psi_b$, then $\psi_- = 0$. Thus **no two identical fermions can occupy the same single-particle state**. This is the Pauli exclusion principle.

1.4 Exchange Forces

For two particles in orthogonal states ψ_a, ψ_b , the expectation value of $(x_1 - x_2)^2$ is:

$$\langle (x_1 - x_2)^2 \rangle_{\text{dist}} = \langle x^2 \rangle_a + \langle x^2 \rangle_b - 2\langle x \rangle_a \langle x \rangle_b, \quad (6)$$

$$\langle (x_1 - x_2)^2 \rangle_{\pm} = \langle (x_1 - x_2)^2 \rangle_{\text{dist}} \mp 2|\langle x \rangle_{ab}|^2, \quad (7)$$

where $\langle x \rangle_{ab} = \int x \psi_a^* \psi_b dx$. Bosons (upper sign) are closer together; fermions (lower sign) are farther apart. This **exchange force** is a quantum geometric effect, not a physical force.

2 Atoms

2.1 Helium

For helium ($Z = 2$), the Hamiltonian is:

$$H = \left(-\frac{\hbar^2}{2m} \nabla_1^2 - \frac{2e^2}{4\pi\epsilon_0 r_1} \right) + \left(-\frac{\hbar^2}{2m} \nabla_2^2 - \frac{2e^2}{4\pi\epsilon_0 r_2} \right) + \frac{e^2}{4\pi\epsilon_0 |\mathbf{r}_1 - \mathbf{r}_2|}. \quad (8)$$

Ignoring electron repulsion, the ground state is:

$$\psi_0(\mathbf{r}_1, \mathbf{r}_2) = \frac{8}{\pi a^3} e^{-2(r_1+r_2)/a}, \quad E_0 = -109 \text{ eV}.$$

The actual ground state energy is -78.975 eV (repulsion raises the energy). The spin state must be antisymmetric (singlet) because the spatial part is symmetric.

Excited states: one electron in ground state, one in excited state. Symmetric spatial $\rightarrow$ singlet (parahelium, higher energy). Antisymmetric spatial $\rightarrow$ triplet (orthohelium, lower energy).

2.2 Periodic Table

Electrons fill hydrogenic orbitals (n, l, m) in order of increasing energy, subject to Pauli exclusion (max 2 electrons per orbital, opposite spins). Shell n holds $2n^2$ electrons. The ground state configuration is determined by:

- Within a shell, lower l has lower energy (due to screening).
- Hund's rules: (1) Maximize total spin S ; (2) For given S , maximize total orbital L ; (3) For subshell $\leq$ half-filled, $J = |L - S|$; if $> 1/2$ filled, $J = L + S$.

Spectroscopic notation: $^{2S+1}L_J$, where $L = 0, 1, 2, 3, \dots$ correspond to S, P, D, F, $\dots$

3 Solids

3.1 Free Electron Gas

Electrons in a rectangular box of dimensions l_x, l_y, l_z with infinite walls. Single-particle states:

$$\psi_{n_x n_y n_z} = \sqrt{\frac{8}{V}} \sin\left(\frac{n_x \pi x}{l_x}\right) \sin\left(\frac{n_y \pi y}{l_y}\right) \sin\left(\frac{n_z \pi z}{l_z}\right), \quad E = \frac{\hbar^2 k^2}{2m},$$

where $k_x = n_x \pi / l_x$, etc., and $V = l_x l_y l_z$. Each state occupies volume π^3 / V in k -space.

For Nq free electrons (density $\rho = Nq/V$), the Fermi wave number is:

$$k_F = (3\rho\pi^2)^{1/3}, \quad E_F = \frac{\hbar^2}{2m}(3\rho\pi^2)^{2/3}. \quad (9)$$

Total energy:

$$E_{\text{tot}} = \frac{\hbar^2 k_F^5 V}{10\pi^2 m} = \frac{\hbar^2 (3\pi^2 Nq)^{5/3}}{10\pi^2 m} V^{-2/3}. \quad (10)$$

Degeneracy pressure (from Pauli exclusion) stabilizes cold solids:

$$P = \frac{2}{3} \frac{E_{\text{tot}}}{V} = \frac{(3\pi^2)^{2/3} \hbar^2}{5m} \rho^{5/3}. \quad (11)$$

3.2 Band Theory (Periodic Potentials)

Bloch's theorem: For a periodic potential $V(x+a) = V(x)$, the eigenfunctions satisfy:

$$\psi(x+a) = e^{iKa} \psi(x), \quad (12)$$

with K real. The energy spectrum consists of **bands** separated by **gaps**.

For the Dirac comb (periodic delta wells):

$$\cos(Ka) = \cos(ka) - \beta \frac{\sin(ka)}{ka}, \quad \beta = \frac{m\alpha a}{\hbar^2}, \quad k = \frac{\sqrt{2mE}}{\hbar}. \quad (13)$$

Allowed energies occur where $|\cos(Ka)| \leq 1$. Materials with completely filled bands are insulators; partially filled bands are conductors.

4 Quantum Statistical Mechanics

4.1 Counting States

For a configuration $\{N_n\}$ (occupation numbers for energy level E_n with degeneracy d_n):

- **Distinguishable particles** (Maxwell-Boltzmann):

$$Q = N! \prod_n \frac{d_n^{N_n}}{N_n!}.$$

- **Identical fermions** (Fermi-Dirac):

$$Q = \prod_n \frac{d_n!}{N_n! (d_n - N_n)!}.$$

- **Identical bosons** (Bose-Einstein):

$$Q = \prod_n \frac{(N_n + d_n - 1)!}{N_n! (d_n - 1)!}.$$

4.2 Most Probable Configuration (Lagrange Multipliers)

Maximize $\ln Q$ subject to $\sum N_n = N$ and $\sum N_n E_n = E$. Using Stirling's approximation $\ln(z!) \approx z \ln z - z$:

- Maxwell-Boltzmann: $N_n = d_n e^{-(\alpha + \beta E_n)}$
- Fermi-Dirac: $N_n = \frac{d_n}{e^{\alpha + \beta E_n} + 1}$
- Bose-Einstein: $N_n = \frac{d_n}{e^{\alpha + \beta E_n} - 1}$

4.3 Physical Interpretation

For a free particle in a box, $\beta = 1/(k_B T)$ and $\mu = -\alpha k_B T$ (chemical potential). The occupation number per single-particle state of energy ϵ is:

$$n(\epsilon) = \begin{cases} e^{-(\epsilon - \mu)/k_B T}, & \text{Maxwell-Boltzmann (classical)} \\ \frac{1}{e^{(\epsilon - \mu)/k_B T} + 1}, & \text{Fermi-Dirac (fermions)} \\ \frac{1}{e^{(\epsilon - \mu)/k_B T} - 1}, & \text{Bose-Einstein (bosons)} \end{cases} \quad (14)$$

At $T = 0$, Fermi-Dirac becomes a step function: all states with $\epsilon < \mu(0) = E_F$ filled, all above empty.

4.4 Blackbody Radiation

Photons are massless bosons with $\mu = 0$ and two spin states. Planck's blackbody spectrum:

$$\rho(\omega) = \frac{\hbar \omega^3}{\pi^2 c^3 (e^{\hbar \omega / k_B T} - 1)}. \quad (15)$$

Stefan-Boltzmann law: total energy density $\propto T^4$.

5 Key Results Summary

$$\text{Two-particle wave function: } \psi_{\pm} = \frac{1}{\sqrt{2}} [\psi_a(1)\psi_b(2) \pm \psi_b(1)\psi_a(2)]. \quad (16)$$

$$\text{Pauli exclusion: No two fermions in same state.} \quad (17)$$

$$\text{Fermi energy: } E_F = \frac{\hbar^2}{2m} (3\rho\pi^2)^{2/3}. \quad (18)$$

$$\text{Bloch theorem: } \psi(x + a) = e^{iK a} \psi(x). \quad (19)$$

$$\text{Fermi-Dirac: } n(\epsilon) = \frac{1}{e^{(\epsilon - \mu)/k_B T} + 1}. \quad (20)$$

$$\text{Bose-Einstein: } n(\epsilon) = \frac{1}{e^{(\epsilon - \mu)/k_B T} - 1}. \quad (21)$$

$$\text{Planck spectrum: } \rho(\omega) = \frac{\hbar \omega^3}{\pi^2 c^3 (e^{\hbar \omega / k_B T} - 1)}. \quad (22)$$